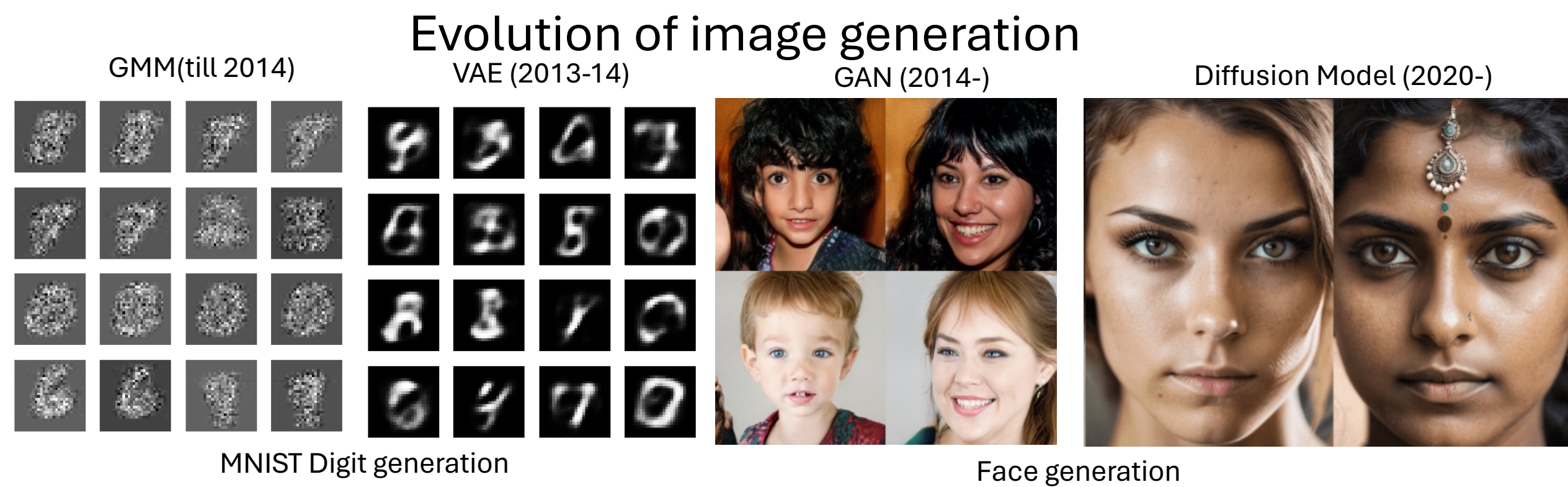


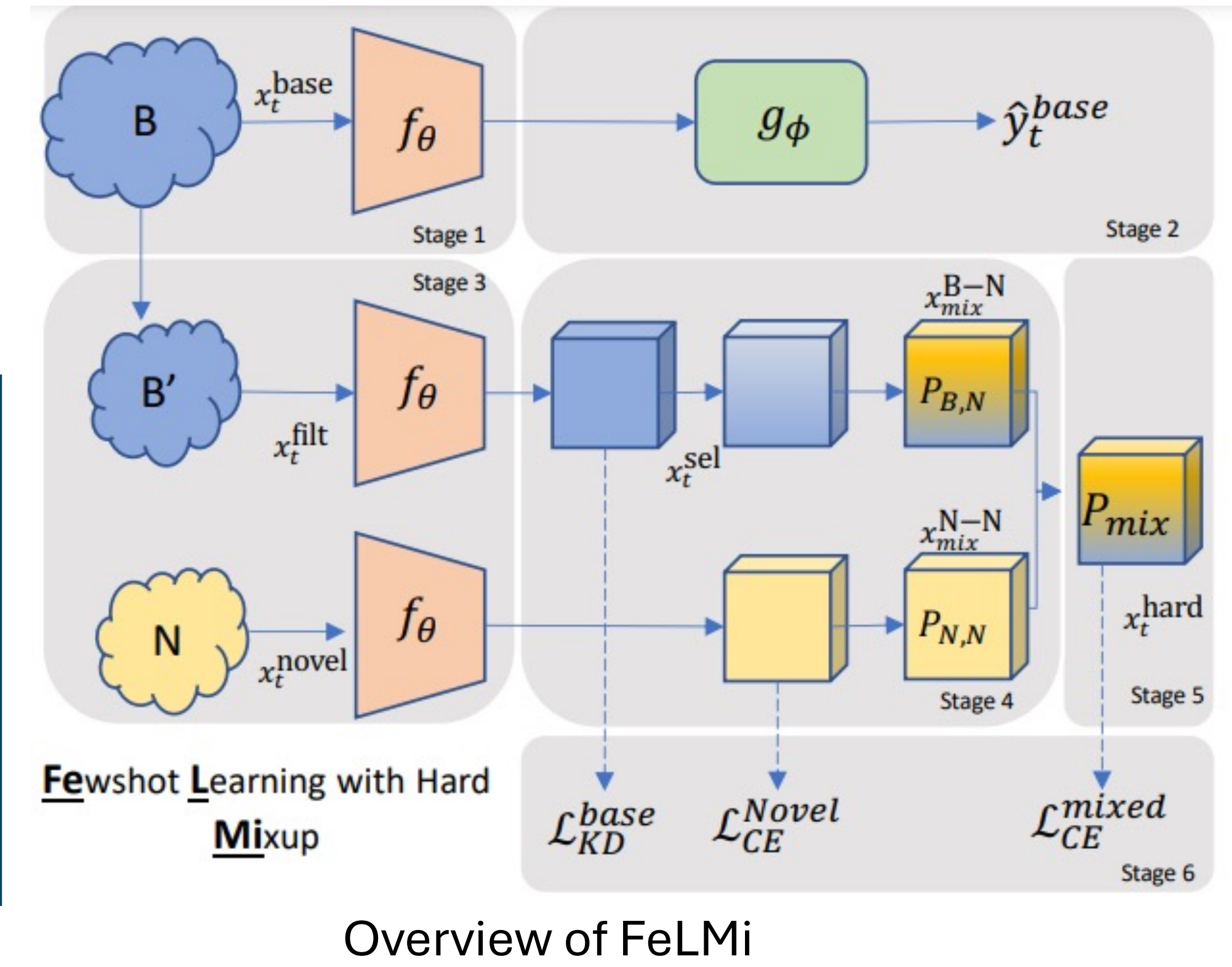
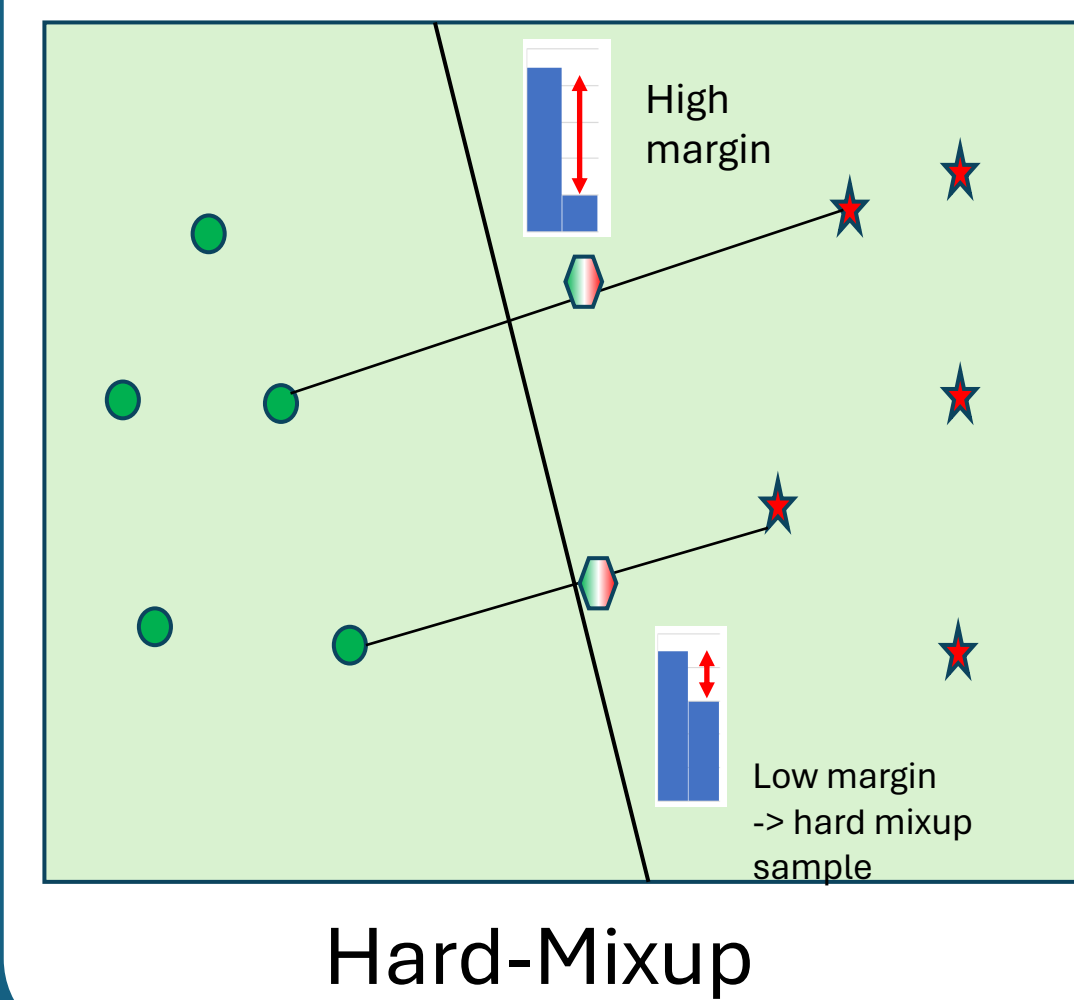
Introduction

- Areas addressed in this work:
 - Few-shot and Multimodal few-shot learning.
 - Generative models for true out-of-distribution tasks, e.g., visual illusion generation and understanding.
 - Improving generation quality using natural image priors and plug-and-play loss formulation.
 - Multi-condition image generation (content-style) using parameter-efficient Low Rank Adapter (LoRA) merging framework.



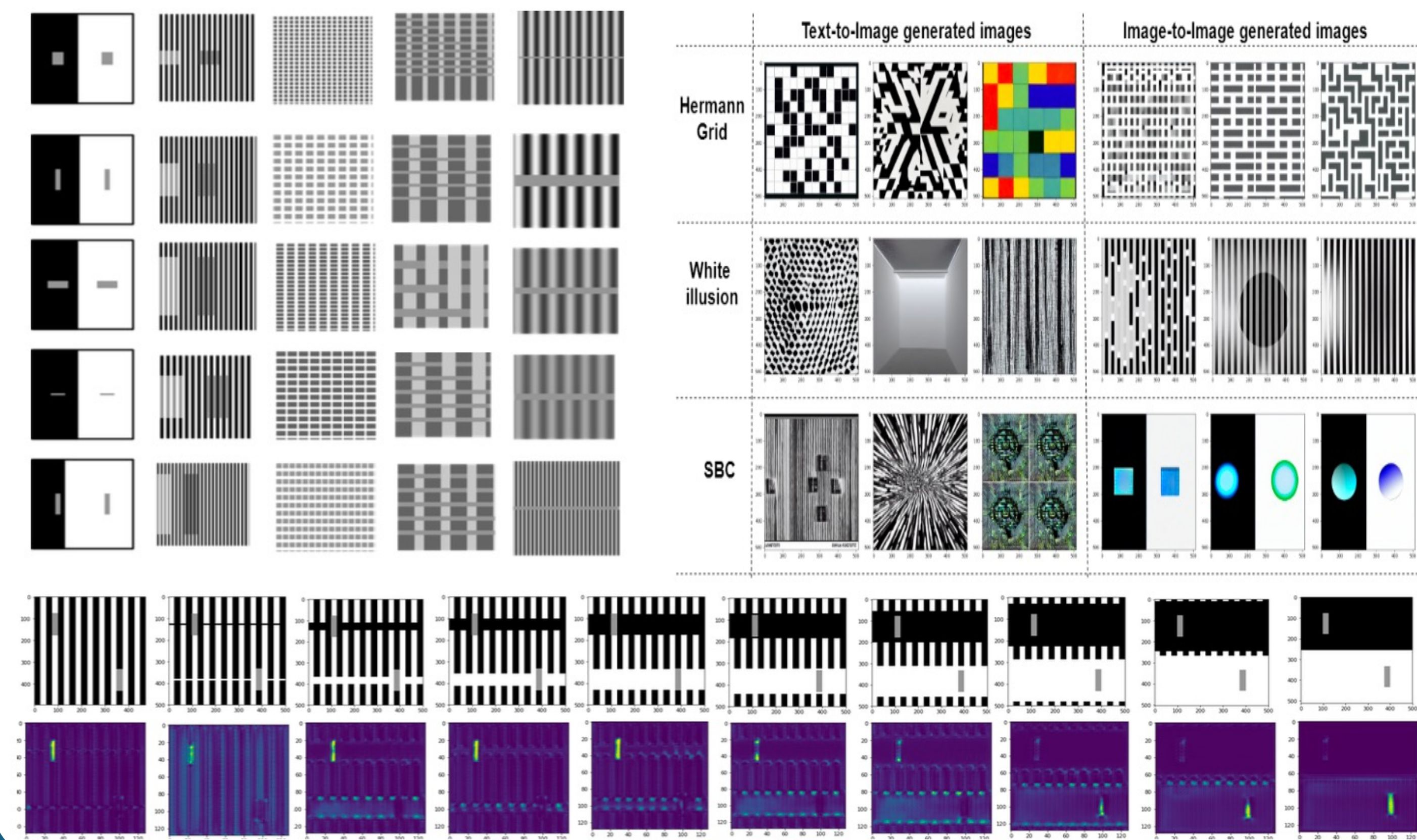
FelMi: Few-shot Learning with Hard-Mixup (NeurIPS'22)

- Propose Hard-mixup data augmentation.
- Improve classification accuracy on few-shot learning benchmarks.



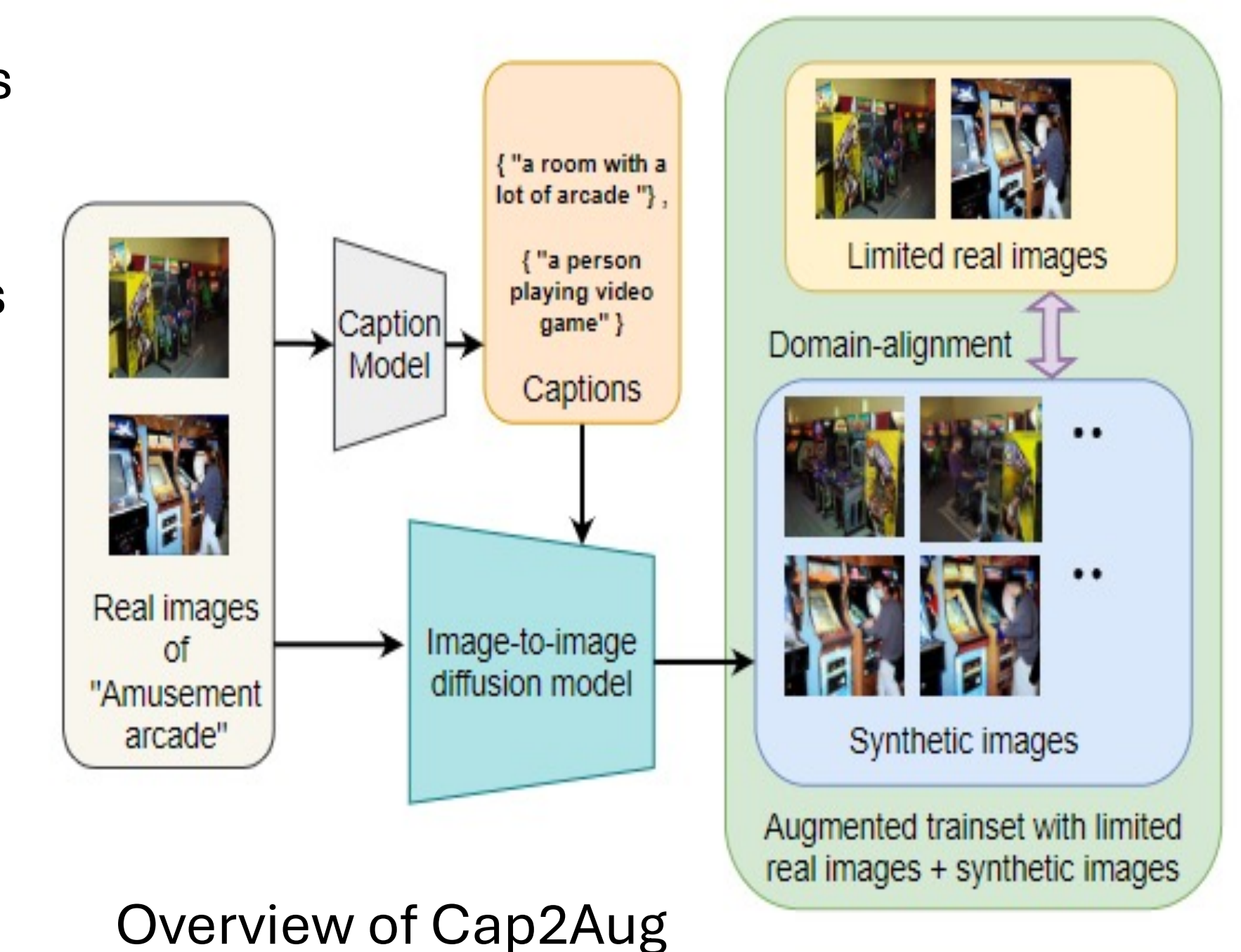
BRI3L: Illusion generation by Generative Diffusion model (ICIP'24)

- We analyze illusory perception of text-to-image (T2I) and image-to-image (I2I) diffusion model.
- We observe T2I models are bad generating illusions, I2I models are better.



Cap2Aug: Caption-Guided Image Data Augmentation (WACV'25)

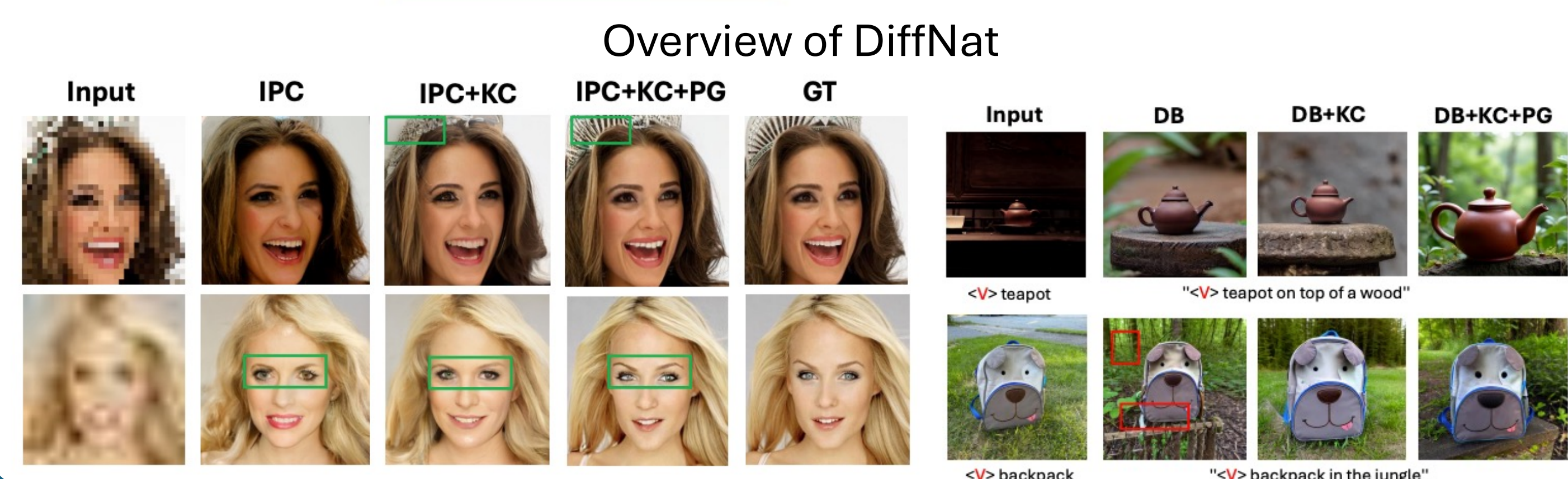
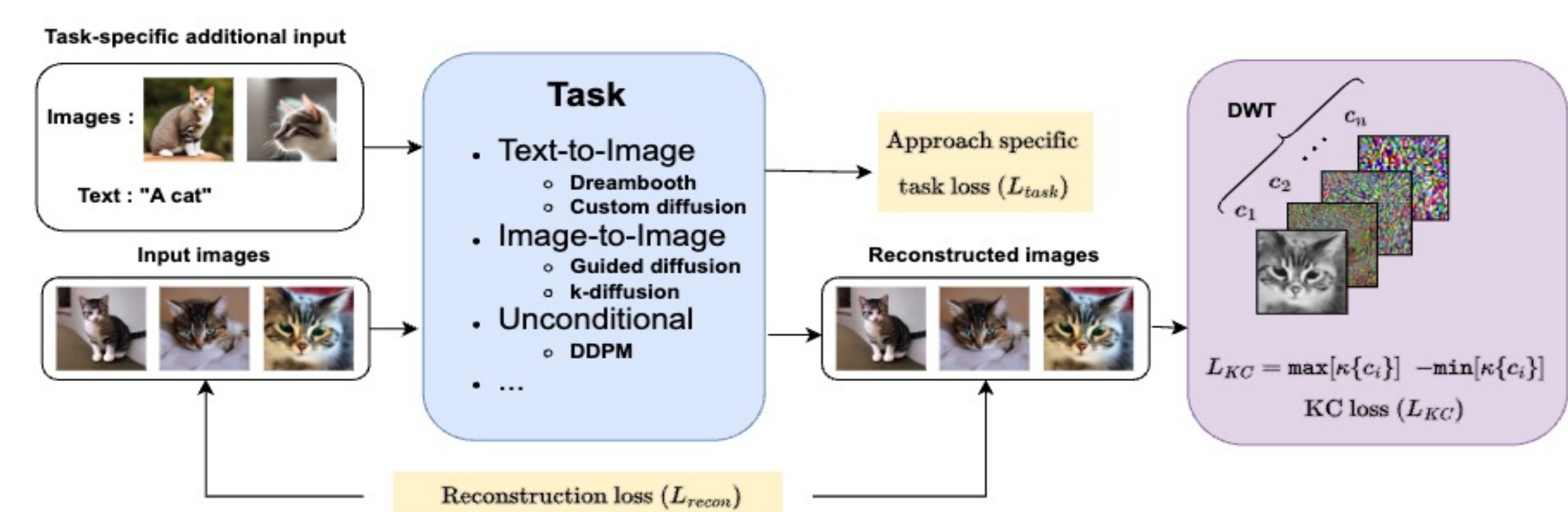
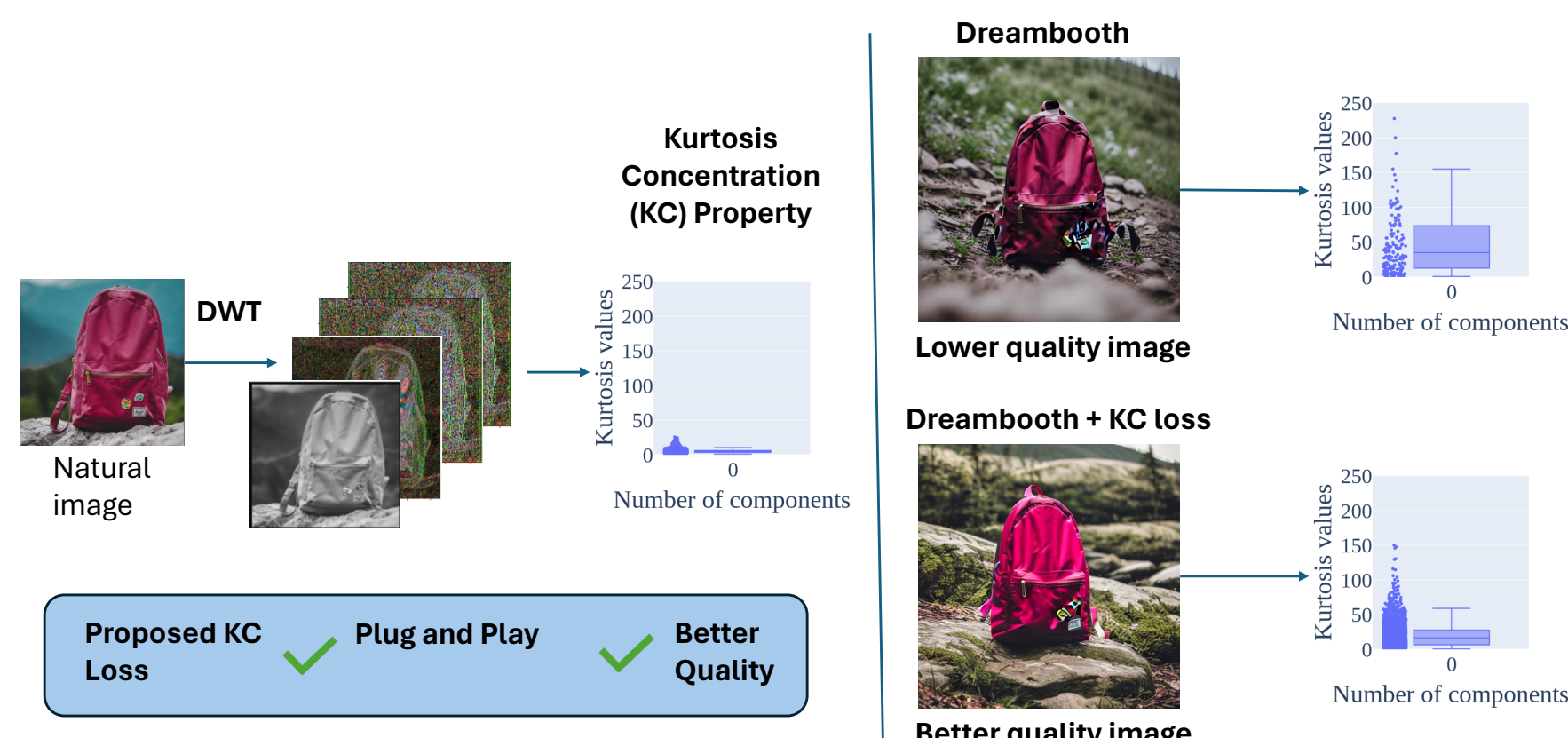
- Idea: Can we use generative models to generate augmentations with semantic diversity?
- Generate captions from the few-examples, use these captions generate augmented version of images using image-to-image diffusion models.



	EuroSAT				SUN397				UCF101			
Shots	2	4	8	16	2	4	8	16	2	4	8	16
Tip [70]	61.68	65.32	67.95	70.50	62.70	64.15	65.62	66.85	64.74	66.46	68.68	70.58
CoOp [72]	61.50	70.18	76.73	82.53	59.48	63.47	65.52	69.26	64.09	67.03	71.92	75.71
Tip-F [70]	66.15	74.12	77.30	82.54	63.64	66.21	68.87	70.47	66.43	70.55	74.01	77.03
Ours	67.03	77.37	77.50	83.64	64.61	67.47	68.93	70.90	68.57	71.76	74.12	77.24
Δ	+0.88	+3.25	+0.20	+1.10	+0.97	+1.26	+0.06	+0.43	+2.14	+1.21	+0.11	+0.21

DiffNat: Improving Diffusion Image Quality using Natural Image Statistics (TMLR'25)

- Use Kurtosis Concentration (KC) Property of natural image.
- Plug-and-play KC loss and Perceptual guidance for improving image quality.
- Validated on text-to-image, image-to-image, super-resolution, personalized few-shot finetuning, video generation tasks.



DuoLoRA: Cycle-consistent and Rank-disentangled Content-Style Personalization (ICCV'25)

- Propose novel rank-dimensional masking providing adaptive rank flexibility.
- Use Diffusion layer-priors and cycle-consistency for better merging.
- Parameter-efficient, reduces learnable params by 19x (SDXL)

